FLUID MECHANICS

Subject Code: 23A01303

UNIT - 2: Fluid Statics

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Fluid pressure: pressure at a point, Pascal's law, pressure variation with temperature, density and altitude. Piezometer, U-tube manometer, single column manometer, U-tube differential manometer. Pressure gauges. Hydrostatic pressure and force on horizontal, vertical and inclined surfaces. Buoyancy and stability of floating bodies.

1. Fluid Pressure

Pressure

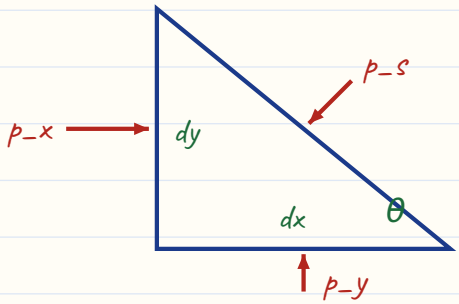
Pressure (intensity of pressure) is the normal force exerted by a fluid per unit area. $p = F/A$. Unit: $N/m^2 = Pa$; 1 kPa = 10^3 Pa; 1 bar = 10^5 Pa.

- Pressure is a scalar; in a fluid at rest it acts normal to any surface (no shear stress when fluid is static).
- 1 atm = 101.325 kPa = 760 mm Hg = 10.33 m of water = 1.01325 bar = 14.7 psi.

Pressure at a Point – Pascal's Law

Pascal's Law

The pressure at a point in a fluid at rest is the same in all directions: $p_x = p_y = p_z$. Any pressure applied to a confined fluid is transmitted undiminished in all directions.



Element is of unit width

Forces (p × area) in x and y:

$p_x\, dy = p_s\, ds \sin\theta$

$p_y\, dx = p_s\, ds \cos\theta + W/2$

Fig 2.1: Small wedge-shaped element in a static fluid

Proof: Take a small wedge of width unity, base dx, height dy, inclined face ds at angle θ. Being at rest, forces balance in x and y (weight is of second order and neglected).

- x-direction: $p_x\, dy = p_s\, ds \sin\theta = p_s\, dy$ (since $ds \sin\theta = dy$) $\rightarrow p_x = p_s$
- y-direction: $p_y\, dx = p_s\, ds \cos\theta + \tfrac{1}{2} \rho g\, dx\, dy = p_s\, dx + (\text{small}) \rightarrow p_y = p_s$ as $dx, dy \rightarrow 0$

- Hence $p_x = p_y = p_s$. The angle θ is arbitrary, so pressure at a point is independent of direction.

Hydraulic Press (Application)



Fig 2.2: Hydraulic lift / press

$$p = F_1/a = W/A \rightarrow W = F_1 \times (A/a)$$

A small force on a small piston creates a large force on a large piston (force multiplication = A/a). Used in hydraulic jacks, brakes, presses and lifts.

Example 1
The diameters of a small and large piston of a hydraulic jack are 5 cm and 25 cm. Find the force needed on the small piston to lift a load of 5000 N.
Sol: $A/a = (25/5)^2 = 25$. $F = W/(A/a) = 5000/25 = 200$ N

2. Hydrostatic Law and Pressure Variation

Consider a vertical fluid column of cross-section dA and height dz in a static fluid (z measured upward). Balancing vertical forces on the element:

$$p\,dA - (p + dp)\,dA - \rho\,g\,dA\,dz = 0 \rightarrow dp/dz = -\rho\,g$$

- This is the basic equation of fluid statics. Pressure increases with depth, and is constant on any horizontal plane in the same continuous fluid.
- Hydrostatic law: The rate of increase of pressure in a vertically downward direction equals the specific weight of the fluid at that point.
- For an incompressible fluid (ρ = const), integrate from the free surface: $p = p_{atm} + \rho\,g\,h$ (absolute), $p_{gauge} = \rho\,g\,h$.
- Pressure head: $h = p/(\rho\,g)$ – the height of a fluid column that produces pressure p. 1 N/cm^2 = 1.02 m of water.

Absolute, Gauge and Vacuum Pressure

Term	Definition	Relation
Absolute pressure	Measured above absolute zero (perfect vacuum)	$p_{abs} = p_{atm} + p_{gauge}$
Gauge pressure	Measured above the local atmospheric pressure	$p_{gauge} = p_{abs} - p_{atm}$
Vacuum (suction) pressure	Pressure below atmospheric, as a positive number	$p_{vac} = p_{atm} - p_{abs}$

Example 2

Find the gauge and absolute pressure at a depth of 10 m in (a) water, (b) oil of $S = 0.8$. Take $p_{atm} = 101.3$ kPa.

Sol: (a) $p_g = \rho g h = 1000 \times 9.81 \times 10 = 98100$ Pa $= 98.1$ kPa; $p_{abs} = 98.1 + 101.3 = 199.4$ kPa

(b) $p_g = 800 \times 9.81 \times 10 = 78480$ Pa $= 78.5$ kPa; $p_{abs} = 78.5 + 101.3 = 179.8$ kPa

Equivalent head of water for oil column $= S \times h = 0.8 \times 10 = 8$ m of water.

Example 2(b) – Pressure in a layered liquid and unit conversion

A tank holds 2 m of oil ($S = 0.8$) floating on 3 m of water. Find the gauge pressure at the bottom. Also express a pressure of 2.5 bar in metres of water and mm of mercury.

Sol: $p = \rho_{oil} g h_1 + \rho_w g h_2 = 800 \times 9.81 \times 2 + 1000 \times 9.81 \times 3 = 15696 + 29430 = 45126$ Pa $= 45.13$ kPa

2.5 bar $= 2.5 \times 10^5$ Pa: $h_w = p/(\rho_w g) = 250000/9810 = 25.48$ m of water; $h_{Hg} = 250000/(13600 \times 9.81) = 1.874$ m $= 1874$ mm Hg

Variation of Pressure with Altitude, Density and Temperature (Compressible Fluids)

For gases the density changes with pressure and temperature. Using the perfect-gas law $p = \rho R T$ in $dp/dz = -\rho g$:

$$dp/p = -(g/RT)\,dz$$

- Isothermal atmosphere ($T = T_0$ constant): $p = p_0 \exp[-g(z - z_0)/(R T_0)]$ – pressure falls exponentially with altitude.
- Linear temperature variation (troposphere): $T = T_0 - \beta z$, β = lapse rate = 0.0065 K/m up to 11 km. Then
  – $p/p_0 = (1 - \beta z/T_0)^{g/(\beta R)}$ (exponent $g/\beta R = 5.26$ for air)

- Adiabatic (isentropic) atmosphere: $p/\rho^k = \text{const}$, giving $T/T_0 = (p/p_0)^{(k-1)/k}$.
- Density variation: for liquids ρ is almost constant, so p varies linearly with depth. For gases, p and ρ fall with altitude; at constant p, density falls as T rises ($\rho = p/RT$).

Example 3

Find the pressure at an altitude of 3000 m in a standard atmosphere ($T_0 = 288.15$ K, $p_0 = 101.325$ kPa, $\beta = 0.0065$ K/m, $R = 287$ J/kg·K).

Sol: $1 - \beta z/T_0 = 1 - (0.0065 \times 3000)/288.15 = 0.9323$; $g/(\beta R) = 9.81/(0.0065 \times 287) = 5.259$

$p = 101.325 \times (0.9323)^{5.259} = 101.325 \times 0.6917 = 70.1$ kPa

3. Measurement of Pressure – Manometers

Manometers are devices that measure pressure by balancing it against a column of liquid. Gauge pressure is found by using the hydrostatic law.

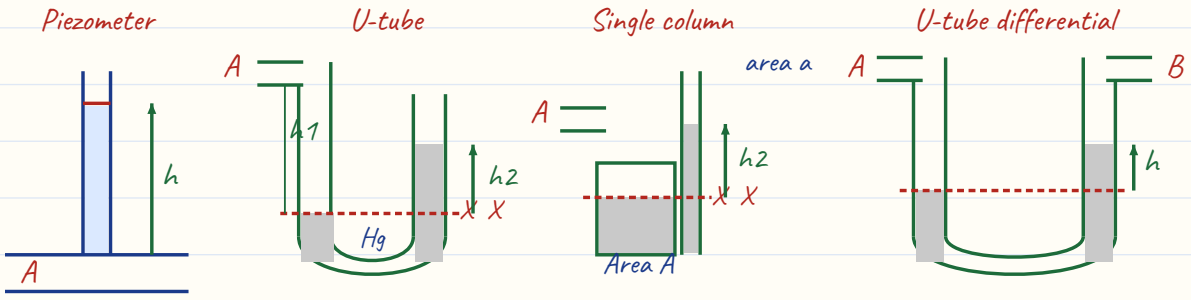


Fig 2.3: Types of manometers

Rules for Solving Manometer Problems

1. Choose a datum line (X–X) through the lowest common surface of the same liquid.
2. Pressures at the same level in the same continuous liquid are equal.
3. Moving down in a liquid → add $\rho g h$; moving up → subtract $\rho g h$.
4. Start from one end and write the pressure up to the other end (open to atmosphere → gauge pressure zero).

(a) Piezometer

The simplest manometer: a vertical glass tube open at the top, fixed to the point where pressure is required.

$$p_A = \rho g h$$

- Only for moderate gauge pressures of liquids; cannot measure gas pressure (gas escapes) or large pressures (tube too tall), nor negative pressure.

- Tube bore should be > 12 mm to avoid capillary error.

(b) U-Tube Manometer

A U-shaped glass tube holds a heavy manometric liquid (mercury, $S = 13.6$) which is immiscible with the fluid whose pressure is to be measured; one end connected to the point, the other open to atmosphere.

$$p_A + \rho_1 g h_1 = \rho_2 g h_2 \rightarrow p_A = \rho_2 g h_2 - \rho_1 g h_1$$

- ρ_1 = density of the fluid in pipe, ρ_2 = density of the manometric liquid, h_1 = height of pipe fluid above X–X, h_2 = height of manometric liquid above X–X in the open limb.
- For a gas in pipe, $\rho_1 g h_1$ is negligible: $p_A = \rho_2 g h_2$. For vacuum (negative) pressure the mercury is pulled up on the pipe side and $p_A = -\rho_2 g h_2 - \rho_1 g h_1$.

Example 4

A U-tube manometer connected to a pipe containing oil of $S = 0.9$ shows mercury ($S = 13.6$) 200 mm above the datum in the open limb. The oil column above the datum in the pipe limb is 100 mm. Find the gauge pressure in the pipe.

Sol: $p_A + \rho_{oil} g (0.1) = \rho_{Hg} g (0.2)$

$p_A = 13600 \times 9.81 \times 0.2 - 900 \times 9.81 \times 0.1 = 26683 - 883 = 25800\ N/m^2 = 25.8\ kPa$ (gauge)

(c) Single Column Manometer (Micromanometer type)

A modification of the U-tube with a large reservoir (area A) on one side and a narrow tube (area a) on the other. Because $A \gg a$, the level fall in the reservoir is very small, so only the rise h_2 in the narrow limb is read – more sensitive.

When pressure p_A is applied to the reservoir, mercury falls Δh in the reservoir and rises h_2 in the narrow limb. Volume balance: $A \Delta h = a h_2 \rightarrow \Delta h = (a/A) h_2$.

$$p_A + \rho_1 g (h_1 + \Delta h) = \rho_2 g (h_2 + \Delta h)$$

$$p_A = \rho_2 g h_2 (1 + a/A) - \rho_1 g h_1 (1 + \ldots) \approx \rho_2 g (h_2 + \Delta h) - \rho_1 g (h_1 + \Delta h)$$

- If a/A is very small: $p_A \approx \rho_2 g h_2 - \rho_1 g h_1$. The inclined-tube version further magnifies the reading ($h_2 = l \sin\theta$).

Example 5

A single column manometer is connected to a water pipe. Reservoir/tube area ratio A/a = 100; h_1 = 0.2 m (water above datum), h_2 = 0.5 m (mercury rise in the narrow limb). Find p_A.

Sol: $\Delta h = (a/A)\, h_2 = 0.5/100 = 0.005$ m

$p_A = \rho_{Hg}\, g\, (h_2 + \Delta h) - \rho_w\, g\, (h_1 + \Delta h) = 13600 \times 9.81 \times 0.505 - 1000 \times 9.81 \times 0.205$

$= 67375 - 2011 = 65364\ N/m^2 = 65.4$ kPa

(d) U-Tube Differential Manometer

Measures the difference of pressure between two points A and B in the same pipe or in two different pipes. Both ends are connected to the points; the U-tube holds a heavy liquid.

$$p_A + \rho_1 g\,(h + y) = p_B + \rho_1 g\, y + \rho_m g\, h$$

$$p_A - p_B = g\, h\, (\rho_m - \rho_1)$$ (A, B at same level, same fluid ρ_1)

- If B is lower than A or the fluids differ, write the equation separately for each limb (add/subtract heads as per the rules).
- Inverted U-tube differential manometer uses a light liquid (air or oil) on top for measuring small pressure differences in liquids: $p_A - p_B = \rho_1 g\, h_1 - \rho_s g\, h_s - \rho_2 g\, h_2$ (heads in respective legs).

Example 6

Two pipes A and B at the same level carry water. A differential U-tube manometer containing mercury shows a deflection of 0.3 m. Find $p_A - p_B$.

Sol: $p_A - p_B = g\, h\, (\rho_m - \rho_w) = 9.81 \times 0.3 \times (13600 - 1000) = 9.81 \times 0.3 \times 12600$

$= 37082\ N/m^2 = 37.08$ kPa (head difference = 3.78 m of water)

Mechanical Pressure Gauges

Manometers are not suitable for very high pressures or fluctuating pressures, so mechanical gauges are used.

- Bourdon tube gauge: a curved, flattened elliptical metal tube closed at one end. Pressure inside tends to straighten the tube; the small movement is magnified through a link, sector and pinion to a pointer on a dial. Measures gauge pressure up to ~1000 bar.
- Diaphragm gauge: pressure deflects a thin flexible diaphragm; used for low pressures and corrosive fluids.
- Dead-weight pressure gauge: piston loaded with known weights; used to calibrate other gauges ($p = W/A$).

- Bellows gauge – low pressures; Vacuum gauge – measures pressures below atmosphere; Pressure transducers – electrical (strain gauge, piezoelectric).

4. Hydrostatic Forces on Submerged Surfaces

A fluid at rest exerts a normal force on any submerged surface. We need (i) the total pressure (magnitude of force) and (ii) the centre of pressure (point of application).

(a) Horizontal Plane Surface

Pressure is uniform over the surface (all points at the same depth h):

$$F = p\,A = \rho\,g\,h\,A \text{ acts at the centroid (centre of pressure = centroid)}$$

(b) Vertical and Inclined Plane Surfaces

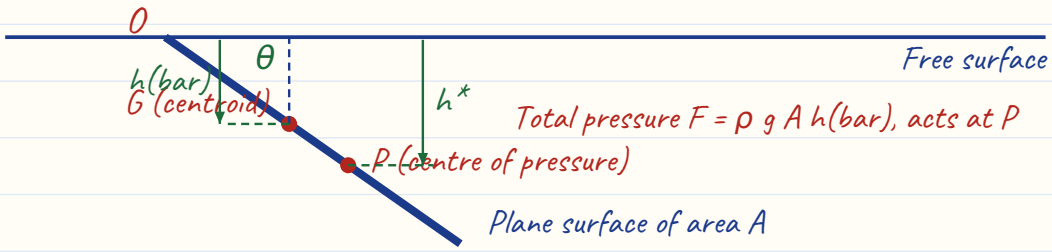


Fig 2.4: Inclined plane surface submerged in a liquid

Let the plane be inclined at angle θ to the free surface, A = area, h(bar) = depth of centroid G below the free surface; y = distance measured along the plane from O (the intersection with free surface). Pressure on a strip dA at depth $h = y\sin\theta$ is $\rho\,g\,y\sin\theta$.

- Total pressure: $F = \int \rho\,g\,y\sin\theta\,dA = \rho\,g\sin\theta \int y\,dA = \rho\,g\sin\theta \cdot A\,y(bar) = \rho\,g\,A\,h(bar)$
- Hence total force = (specific weight) × (area) × (depth of centroid). It is independent of the angle θ for a given centroid depth.

Centre of pressure (h*): Take moments of the elemental forces about O:

$$F \cdot y^* = \int \rho\,g\,y^2\sin\theta\,dA = \rho\,g\sin\theta \cdot I_O$$

using parallel axis theorem $I_O = I_G + A\,y_c^2 \rightarrow$

$$h^* = h(bar) + (I_G \sin^2\theta) / (A\,h(bar))$$

- I_G = second moment of area about the horizontal axis through the centroid. For a vertical surface (θ = 90°): $h^* = h(bar) + I_G/(A\,h(bar))$.
- The centre of pressure is always below the centroid ($h^* > h(bar)$) because pressure increases with depth; it approaches G as the depth increases.

- Horizontal plane: $\sin\theta = 0$, so $h^* = h(\text{bar})$ (coincide).

Shape	Area A	I_G (about horizontal axis through centroid)	Centroid
Rectangle (b wide, d deep)	$b\,d$	$b\,d^3/12$	$d/2$ from top
Triangle (base b, height d)	$\frac{1}{2}\,b\,d$	$b\,d^3/36$	$d/3$ from base
Circle (dia d)	$\pi\,d^2/4$	$\pi\,d^4/64$	centre
Semicircle (radius r)	$\pi\,r^2/2$	$0.1098\,r^4$	$4r/3\pi$ from base

Example 7 – Vertical rectangular gate

A rectangular gate 2 m wide and 3 m high is vertical in a water tank, its top edge at the free surface. Find the total force and depth of centre of pressure.

Sol: $A = 2 \times 3 = 6\ m^2$; $h(\text{bar}) = 1.5\ m$

$F = \rho\,g\,A\,h(\text{bar}) = 9810 \times 6 \times 1.5 = 88290\ N = 88.3\ kN$

$I_G = b\,d^3/12 = 2 \times 27/12 = 4.5\ m^4$; $h^* = h(\text{bar}) + I_G/(A\,h(\text{bar})) = 1.5 + 4.5/(6 \times 1.5) = 2.0\ m$ (= 2/3 of depth, from the free surface)

Example 8 – Inclined circular plate

A circular plate of diameter 2 m is inclined at 30° to the horizontal. Its centre is 4 m below the free surface of water. Find F and the centre of pressure.

Sol: $A = \pi(2)^2/4 = 3.1416\ m^2$; $h(\text{bar}) = 4\ m$

$F = \rho\,g\,A\,h(\text{bar}) = 9810 \times 3.1416 \times 4 = 123276\ N = 123.3\ kN$

$I_G = \pi\,d^4/64 = \pi \times 16/64 = 0.7854\ m^4$; $h^* = 4 + (0.7854 \times \sin^2 30°)/(3.1416 \times 4) = 4 + 0.7854 \times 0.25/12.566 = 4.0156\ m$

Distance along the plate from the free surface $= h^*/\sin\theta = 4.0156/0.5 = 8.03\ m$

Example 8(b) – Vertical triangular plate

A triangular plate of base 3 m and height 2 m is submerged vertically in water with its base horizontal at the free surface and apex downward. Find the force and centre of pressure.

Sol: $A = \frac{1}{2} \times 3 \times 2 = 3\ m^2$; centroid is $d/3 = 0.667\ m$ below the base (free surface): $h(\text{bar}) = 0.667\ m$

$F = \rho\,g\,A\,h(\text{bar}) = 9810 \times 3 \times 0.667 = 19620\ N = 19.62\ kN$

$I_G = b\,d^3/36 = 3 \times 8/36 = 0.667\ m^4$; $h^* = 0.667 + 0.667/(3 \times 0.667) = 1.0\ m$ below the free surface

Remarks

- Pressure diagram (prism): For a rectangular vertical surface, the load is a triangular prism; F acts at its centroid, 2/3 depth from the top.
- If there is liquid on both sides, the net force is the difference of forces and the centre of pressure is found by moments.
- For a curved surface, find horizontal component F_H = force on vertical projection and vertical component F_V = weight of liquid above the surface; resultant $F = \sqrt{(F_H^2 + F_V^2)}$.
- Applications: design of dams, sluice gates, tank walls, lock gates.

5. Buoyancy and Stability of Floating Bodies

Archimedes' Principle
A body wholly or partly immersed in a fluid experiences an upward force (buoyant force) equal to the weight of the fluid displaced by the body.

$$F_B = \rho_{fluid}\, g\, V_{displaced}$$

- Centre of buoyancy (B): the centroid of the displaced volume of fluid; the buoyant force acts through B.
- Proof: upward force on the bottom of a prism = $\rho\, g\, (h + z)\, A$; downward on top = $\rho\, g\, h\, A$; net = $\rho\, g\, z\, A$ = weight of displaced fluid.
- Conditions: $W > F_B$ → body sinks; $W = F_B$ → floats (neutrally buoyant if fully submerged); $W < F_B$ → rises until it floats, with $F_B = W$.
- Law of floatation: A floating body displaces its own weight of fluid; $\rho_b\, V = \rho_f\, V_d$
  Fraction submerged = S_b / S_f
- Applications: ships, submarines, hydrometers (density), balloons, life jackets.

Stability of Submerged and Floating Bodies

- Submerged body – the weight acts at the centre of gravity G and the buoyancy at B. Stable if B is above G (restoring couple on tilting); unstable if B is below G; neutral if B and G coincide.
- Floating body – on tilting, the shape of the displaced volume changes, so B moves to a new position B'. The vertical line through B' cuts the original axis at the metacentre M.

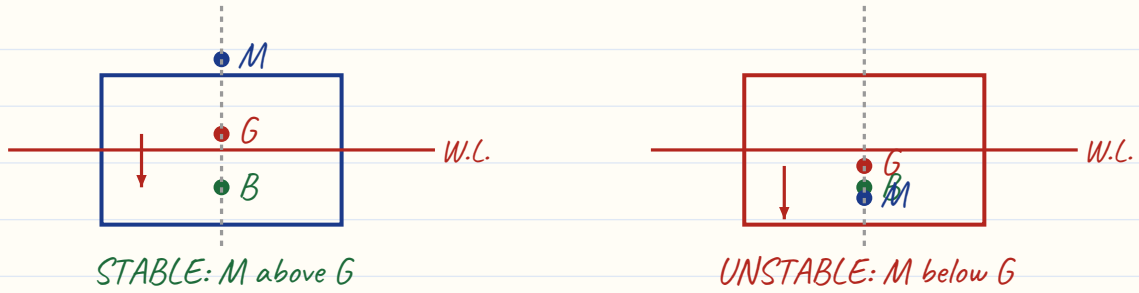


Fig 2.5: Equilibrium conditions of a floating body (B = centre of buoyancy, G = centre of gravity, M = metacentre)

Metacentre (M) and Metacentric Height (GM)

The metacentre is the point about which a floating body starts oscillating when given a small angular displacement; it is the intersection of the line of action of buoyancy (in the displaced position) with the original vertical axis. Metacentric height GM is the distance between G and M.

$$BM = I / V$$

$$GM = BM - BG = I/V - BG$$

- I = second moment of area of the waterline (floating) section about the longitudinal axis (axis of tilt); V = volume of fluid displaced; BG = distance between B and G (positive if G above B).
- Stable equilibrium: M above G ($GM > 0$) – the couple $W \times GM \sin\theta$ restores the body.
- Unstable: M below G ($GM < 0$) – the couple overturns the body. Neutral: M coincides with G ($GM = 0$).
- Righting moment = $W \cdot GM \cdot \sin\theta \approx W \cdot GM \cdot \theta$ (for small angles).
- Time period of rolling: $T = 2\pi \sqrt{(k^2 / (g \cdot GM))}$, k = radius of gyration about the axis of roll.

Experimental Determination of Metacentric Height

A pontoon of weight W floats in water. A movable weight w is placed at the centre and then shifted sideways by a distance x; the angle of heel θ is measured with a plumb-line.

$$GM = (w \cdot x) / (W \cdot \tan\theta)$$ (W = total weight including w)

Example 9 – Metacentric height of a rectangular block

A rectangular block 1.5 m wide, 0.8 m deep and 4 m long has $S = 0.7$ and floats in water with its 4 m side horizontal along the axis. Check its stability for rolling about the longitudinal axis.

Sol: Draught $d = S \times D = 0.7 \times 0.8 = 0.56$ m. G above base $= D/2 = 0.4$ m; B above base $= d/2 = 0.28$ m; $BG = 0.12$ m

$I = L B^3/12 = 4 \times 1.5^3/12 = 1.125\ m^4$; $V = 4 \times 1.5 \times 0.56 = 3.36\ m^3$

$BM = I/V = 0.3348$ m; $GM = BM - BG = 0.3348 - 0.12 = 0.215$ m

GM positive → M above G → the block is stable.

Example 10 – Buoyancy

A wooden block of $S = 0.6$ floats in water. What percentage of its volume is submerged? If a body of 0.5 m^3 is fully immersed in water, what is the buoyant force?

Sol: Fraction submerged $= S_b/S_f = 0.6 \rightarrow 60\ \%$. $F_B = \rho g V = 1000 \times 9.81 \times 0.5 = 4905\ N$

Example 11 – Density of a stone by buoyancy

A stone weighs 392.4 N in air and 250 N when fully immersed in water. Find its volume, density and specific gravity.

Sol: Buoyant force $= 392.4 - 250 = 142.4\ N = \rho_w g V \rightarrow V = 142.4/9810 = 0.01452\ m^3$

Mass $= 392.4/9.81 = 40$ kg; $\rho = 40/0.01452 = 2756\ kg/m^3$; $S = 2.76$

Stability – Points to Remember

- For wide, shallow vessels (large waterline I) BM is large → good stability (barges, pontoons). Narrow, deep bodies tend to roll over.
- Stability can be improved by lowering G (ballast at bottom) or increasing the beam (width) of the vessel.
- A cylinder or sphere floating with its axis vertical must be checked for stability: for a floating sphere, GM depends on the density ratio.
- Ship designers use the metacentric height GM of about 0.3 – 1.5 m; too large a GM makes rolling quick and uncomfortable (stiff ship), too small makes the ship tender and dangerous.

6. Quick Revision Sheet

Concept	Formula
Pascal's law	$p_x = p_y = p_z$; $W = F(A/a)$
Hydrostatic law	$dp/dz = -\rho g$; $p = \rho g h$
Absolute pressure	$p_{abs} = p_{atm} + p_{gauge}$
U-tube manometer	$p_A = \rho_2 g h_2 - \rho_1 g h_1$
Differential manometer	$p_A - p_B = g h (\rho_m - \rho_1)$
Total pressure on plane	$F = \rho g A h(bar)$
Centre of pressure	$h^* = h(bar) + I_G \sin^2\theta / (A h(bar))$
Buoyancy	$F_B = \rho g V_d$
Metacentric height	$GM = I/V - BG$ (stable if $GM > 0$)

Important Questions

1. State and prove Pascal's law.
2. Derive the hydrostatic law and obtain pressure variation in an isothermal and a linear-temperature atmosphere.
3. Compare absolute, gauge and vacuum pressure; explain the working of a Bourdon tube gauge.
4. Explain piezometer, U-tube manometer, single-column manometer and differential manometer with derivations.
5. Derive expressions for total pressure and centre of pressure on an inclined plane surface.
6. Explain buoyancy, centre of buoyancy, metacentre and conditions of stability of a floating body.